The concept of food chains—the idea that all living things are linked through their dependence on other species for their food—dates back many centuries, but it was not until the early 20th century that scientists developed the concept of food chains forming a food web.

The pioneer of this thinking was British zoologist Charles Elton, whose book *Animal Ecology* (1927) describes what he called the “food cycle”. He later went on to develop theories that encompassed more complex interactions between animals and the environment—insights that underpin modern animal ecology. He likened our knowledge of individual plant and animal species to the cells in a beehive—each “cell” of knowledge is important in its own right, but by putting them all together something much more than the sum of the parts is created—the “beehive” of ecology.

Nowadays, the study of animal ecology focuses on how animals interact with their environment, the roles played by different species, why populations rise and fall, why animal behavior sometimes changes, and the impact of environmental change on animals. The principle underlying the work of animal ecologists is that there is generally a balance in nature, so if the population of a given species grows too large it will be regulated, most often by a lack of food. However, relationships between organisms and their environment change from place to place and through time.